SAGAR LAL

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with an MSc in Artificial Intelligence from the University of Verona, completed in October 2026, and three years of professional software development. Thesis work in causal discovery and explainable AI, evaluated across robotics, biomedical and industrial process data. Comfortable from research through to production: Python data pipelines, deep learning on time series, REST APIs and full-stack delivery.

EDUCATION

MSc in Artificial Intelligence (LM-18) 105/110 · October 2026
University of Verona, Italy

- **Thesis:** *Causal Anomaly Detection from Multivariate Time Series* — supervisor Prof. Daniele Meli. Evaluated three causal discovery algorithms (PCMCI, GES, CAM-UV) across 27 configurations on three domains, detecting faults from drift in learned regression coefficients rather than prediction error.
- **Highest marks:** Machine Learning & Deep Learning 30 cum laude, Knowledge Representation 30 cum laude, Logic in AI 30/30.
- **Coursework:** Explainable AI, Computer Vision & Deep Learning, Natural Language Processing, Reinforcement Learning, Planning & Automated Reasoning, Cyber-security for IoT, Biomedical Decision Support Systems.

BSc in Computer Systems Engineering Jan 2019 – Nov 2022
Dawood University of Engineering & Technology, Karachi GPA 3.55 / 4.0

- **Final-year project:** *Automatic Robot Palletization Arm*, as team lead. Built a 3-DoF robotic arm and conveyor that palletize bottles, with IR detection and teach-and-replay control (Arduino Mega, C++, I2C servo driver).

AI AND MACHINE LEARNING EXPERIENCE

Project Manager May 2024 – Dec 2025
Codesy — Remote

- Led cross-functional delivery of Python-based AI automation tools, from requirements through to deployment.
- Oversaw design and deployment of FastAPI REST services backed by PostgreSQL.
- Coordinated data engineering work, automating repetitive processing across datasets of 20,000+ records.

Python · FastAPI · PostgreSQL · Agile delivery

Machine Learning Intern Apr 2025 – Jun 2025
University of Verona — Verona, Italy

- Fine-tuned pretrained ResNet and VGG architectures in TensorFlow for image classification.
- Built preprocessing and augmentation pipelines with OpenCV and PIL to improve robustness.
- Developed LSTM and GRU models for time-series forecasting.

TensorFlow · OpenCV · CNNs · LSTM / GRU

SOFTWARE AND WEB DEVELOPMENT EXPERIENCE

Web Developer Jan 2026 – Present
La Vertueuse — France (Remote)

- Maintain the company's WordPress website and administrative portal, keeping availability high.
- Manage backend systems and extend custom PHP features for content management.
- Integrated Stripe and SumUp payment gateways for e-commerce transactions.

WordPress · PHP · JavaScript · SQL · Stripe API

Freelance Web Developer Mar 2023 – Present
Independent clients — Remote

- Designed and built responsive sites in WordPress and Elementor to client specifications.

WordPress · Elementor · CSS3

Frontend / Full-Stack Developer 2023 – 2025
Confidential (US SaaS company) — Remote

- Built and maintained frontend and backend features in Vue.js, Angular and Next.js for a product serving 10,000+ monthly active users.
- Integrated user interfaces with backend services and improved page load performance.

Vue.js · Angular · Next.js · REST APIs

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL, PHP
Machine learning: PyTorch, TensorFlow, scikit-learn, CNNs, LSTM/GRU, Transformers, transfer learning
AI research: causal discovery (PCMCI, GES, CAM-UV), explainable AI (SHAP, Grad-CAM), anomaly detection, time-series modelling
Data: NumPy, pandas, Matplotlib, exploratory data analysis, statistical testing (bootstrap, paired tests)
Web and backend: FastAPI, Flask, REST APIs, Vue.js, Angular, Next.js, PostgreSQL
Tools: Git, Docker, Linux, Jupyter, LaTeX

CERTIFICATIONS AND LANGUAGES

Certifications: Google Cybersecurity Certificate; Data Analytics & Business Intelligence; Tools of the Trade: Linux & SQL; E-commerce Management
Languages: English (professional proficiency); Italian (A2, certified by the University of Verona); Urdu and Hindi (native)